

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disorder characterized by motor neuron dysfunction and loss leading to muscle atrophy and eventual death. A hallmark of ALS is the dysregulated excitability of neurons, which can manifest as hyperexcitability early in disease progression or hypoexcitability later in disease progression. Research has shown that this altered excitability occurs due to membrane ion channel dysfunction in motor neurons. In addition, altered levels of GABA, an inhibitory neurotransmitter, and glutamate, an excitatory neurotransmitter, also contribute to altered neuron excitability. ALS was originally thought to be a disorder solely of the motor division of the peripheral nervous system; however, it has been found that multiple body systems are affected and symptoms not related to the motor division, such as cognitive impairment, commonly develop over time.

Researchers hypothesized a relationship between the number of synapses (ie, synaptic density) in certain brain regions (frontal and motor cortexes) and the development of cognitive impairment symptoms in individuals with ALS. To investigate this, they measured synaptic density in the frontal and motor cortexes of patients affected with ALS who also demonstrated cognitive impairment (ALSci). These densities were compared to those of patients with ALS who did not demonstrate cognitive impairment (ALSnci) and also to those of healthy individuals without ALS (Table 1).

Table 1 Frontal Cortex and Motor Cortex Synaptic Densities in ALSci, ALSnci, and Healthy Patients: Means and 95% Confidence Intervals

Patient condition	Frontal cortex synaptic density (synapses/mm ³)	Motor cortex synaptic density (synapses/mm ³)
ALSci	$6.75 \times 10^8 \pm 0.25 \times 10^8$	$7.25 \times 10^8 \pm 0.25 \times 10^8$
ALSnci	$7.50 \times 10^8 \pm 0.75 \times 10^8$	$7.10 \times 10^8 \pm 0.50 \times 10^8$
Healthy	$8.00 \times 10^8 \pm 0.25 \times 10^8$	$7.00 \times 10^8 \pm 0.50 \times 10^8$

Compared to healthy patients, is the development of ALS with cognitive impairment associated with synaptic density in the frontal cortex, motor cortex, or both? A relationship between the two exists in:

- ✔ ☒ A. the frontal cortex only. [56%]
- ☐ B. the motor cortex only. [6%]
- ☐ C. both the frontal and motor cortexes. [28%]
- ☐ D. neither the frontal nor motor cortexes. [8%]

Correct

Explanation:

Relationship between cognitive impairment and synaptic density

Patient condition	Frontal cortex synaptic density (synapses/mm ²)	Motor cortex synaptic density (synapses/mm ²)
ALSci	$6.75 \times 10^8 \pm 0.25 \times 10^8$	6.50×10^8 to 7.00×10^8 $7.25 \times 10^8 \pm 0.25 \times 10^8$
ALSnci	$7.50 \times 10^8 \pm 0.75 \times 10^8$	$7.10 \times 10^8 \pm 0.50 \times 10^8$
Healthy	$8.00 \times 10^8 \pm 0.25 \times 10^8$	$7.00 \times 10^8 \pm 0.50 \times 10^8$ 7.75×10^8 to 8.25×10^8

These ranges do not overlap and are therefore significantly different.

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Synapses are gaps between the presynaptic and postsynaptic neuron, and they can be **electrical or chemical** in nature. These interfaces are important in allowing **communication** between an axon terminal of a presynaptic neuron and the dendrites of a postsynaptic neuron as a result of an action potential.

In the passage, researchers hypothesized a relationship between synaptic density (ie, number of synapses per unit area) in the frontal cortex and motor cortex and the development of cognitive impairment symptoms in individuals with ALS. To study this, they measured synaptic density in both brain regions in patients affected with ALS that demonstrated cognitive impairment (ALSci). These densities were then compared to those in patients with ALS not demonstrating cognitive impairment (ALSnci) and individuals without ALS (healthy).

Table 1 shows that patients with ALSci had a **significantly lower** density of synapses in the frontal cortex compared to healthy patients (ie, without ALS) (**Choice D**). This result indicates that in the frontal cortex, cognitive impairment in ALS is likely related to a *loss* of healthy synapse density.

In contrast, patients with ALSci showed **no significant difference** in motor cortex synaptic density from those of healthy patients (**Choices B and C**). Therefore, a **relationship between the development of cognitive impairment and synaptic density** exists in the **frontal cortex only** (*not* the motor cortex).

Educational objective:

Synapses allow communication between presynaptic neurons and postsynaptic neurons.

Time spent: 0 sec
Subject:
Biology

QID: 403478
System:
Endocrine and Nervous Systems

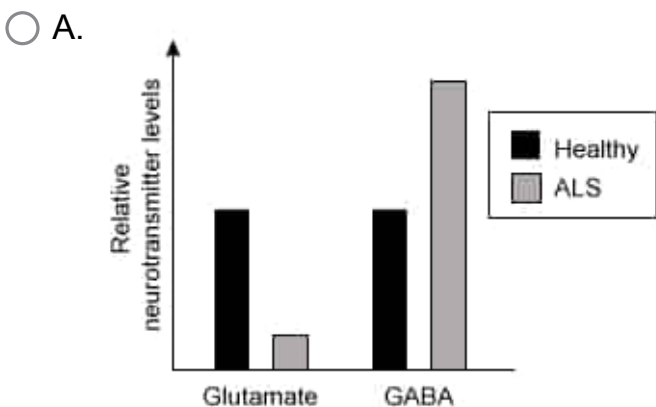
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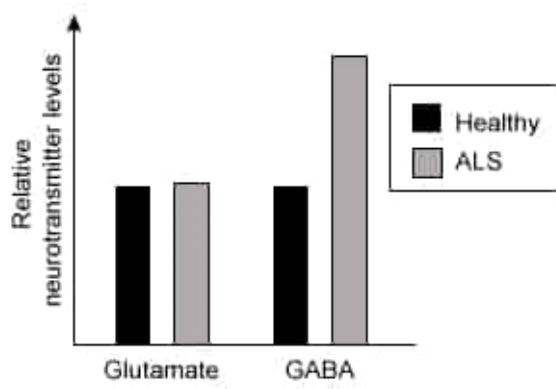
Patient condition	Frontal cortex synaptic density (synapses/mm ³)	Motor cortex synaptic density (synapses/mm ³)
ALSci	$6.75 \times 10^8 \pm 0.25 \times 10^8$	$7.25 \times 10^8 \pm 0.25 \times 10^8$
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To elucidate the effects of brain neurotransmitter levels on ALS progression, relative GABA and glutamate levels were measured in patients recently diagnosed with ALS and compared to age-matched patients without ALS. Which graph shows the expected results of this study?



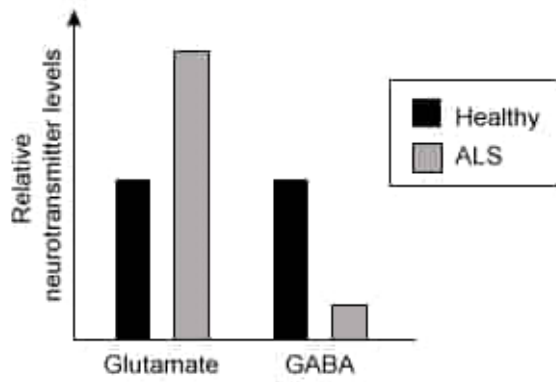
[19%]

○ B.



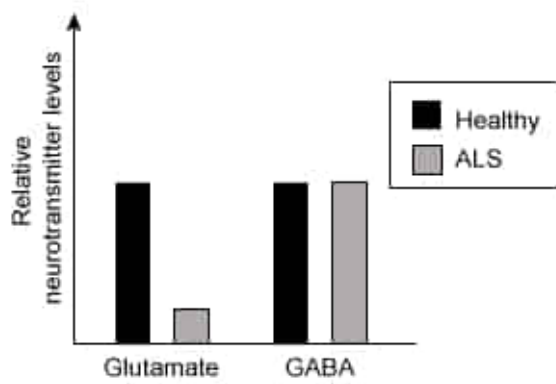
[3%]

✓ C.



[75%]

○ D.

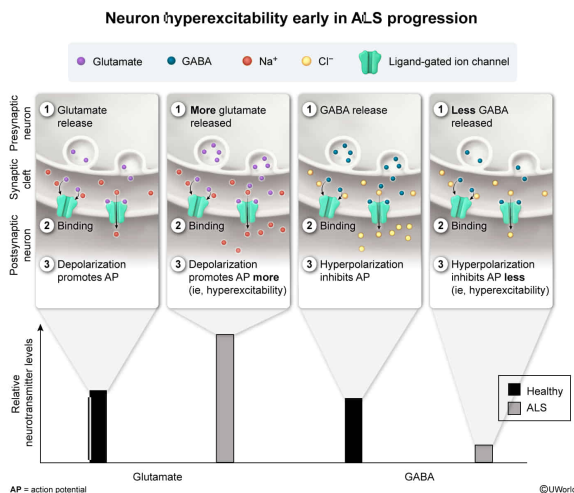


[1%]

Correct

74% answered correctly

Explanation:



Neurons communicate with one another via electrochemical signals known as **action potentials**. Specifically, an action potential **travels** down the signaling cell (**presynaptic neuron**) until it reaches the axon terminals, which are responsible for **signal transmission** across the synapse and to the **postsynaptic neuron**. At **chemical synapses**, an action potential in the presynaptic neuron results in the release of **neurotransmitters**, which bind receptors on the membrane of the postsynaptic neuron.

Depending on their effect on the postsynaptic cell, some neurotransmitters are classified as **excitatory** (ie, they make an action potential *more* likely to occur in the postsynaptic neuron). Other neurotransmitters are classified as **inhibitory** (ie, they make an action potential *less* likely to occur in the postsynaptic neuron).

The passage states that in ALS, excitability of neurons is disrupted. Specifically, *early* in disease progression, neurons display *hyperexcitability* whereas later in disease progression, they display hypoexcitability. The passage also states that this dysfunctional excitability is due partially to altered levels of GABA, an inhibitory neurotransmitter, and glutamate, an excitatory neurotransmitter.

In this question, relative GABA and glutamate levels were measured in patients recently diagnosed with ALS and in age-matched healthy patients to elucidate the effects of brain neurotransmitter levels on ALS progression. Because the experimental patients were *recently diagnosed* with ALS, they would be expected to show neuronal *hyperexcitability*. This result could be due to *lower-than-normal* levels of *inhibitory* neurotransmitters (eg, GABA), *increased* levels of excitatory neurotransmitters (eg, glutamate), or both. The graph shown in Choice C is the only option that correctly depicts this expected result.

In patients with ALS, decreased glutamate and increased GABA **levels (Choice A)**, normal (ie, healthy) glutamate and increased GABA **levels (Choice B)**, and decreased glutamate and normal GABA **levels (Choice D)** would all result in hypoexcitability, *not* hyperexcitability.

Educational objective:

The summation of signals from excitatory neurotransmitters, which make an action potential in a postsynaptic cell more likely, and inhibitory neurotransmitters, which make an action potential in a postsynaptic cell less likely, determine the net effect on the postsynaptic cell.

Time spent:0 sec

Subject:
Biology

QID:403479

System:
Endocrine and Nervous Systems

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Voltage-gated potassium ion channels were found to be downregulated in a patient with ALS. Would the patient's motor neurons be expected to display hypoexcitability or hyperexcitability?

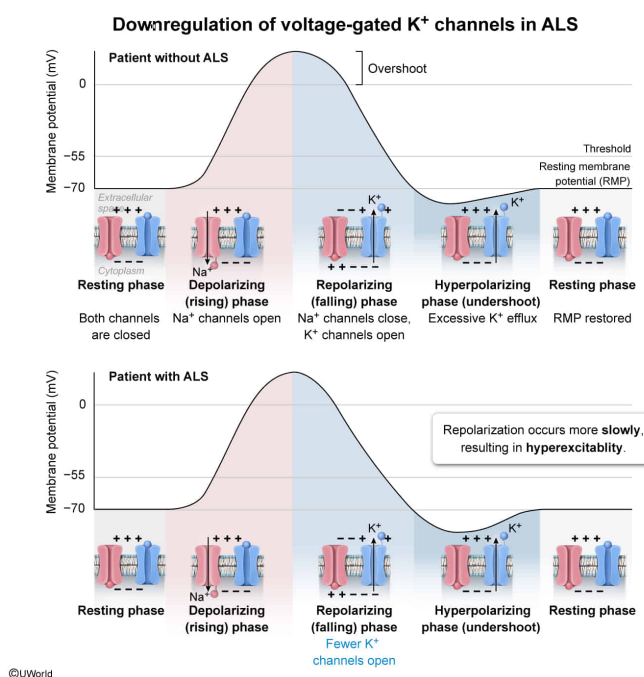
- ☐ A. Hypoexcitability, because depolarization would be more likely in motor neurons. [4%]

- ☐ B. Hypoexcitability, because repolarization of motor neurons would occur more slowly. [40%]
- ☐ C. Hyperexcitability, because depolarization would be less likely in motor neurons. [5%]
- ✓ ☒ D. Hyperexcitability, because repolarization of motor neurons would occur more slowly. [50%]

Correct

50% answered correctly

Explanation:



Action potentials (APs) are electrical signals that **travel** from a **neuron's** cell body, down its axon, and to its axon terminals. During APs, the **membrane potential** of the neuron changes due to opening and closing of voltage-gated sodium (Na⁺) and potassium (K⁺) **ion channels** as follows:

1. At rest, a neuron **maintains** a potential difference known as the **resting membrane potential** (RMP) of approximately -70 mV, meaning that the intracellular space is 70 mV more negative than the extracellular space. The RMP is maintained by the **Na⁺/K⁺ pump** and K⁺ leak channels, or channels that are always open and allow **passive diffusion** of K⁺ across the membrane.
2. If a stimulus causes voltage-gated Na⁺ channels to open, Na⁺ will

rush into the cell, causing the membrane to **depolarize** (ie, the membrane potential becomes more positive). If this depolarization is enough to reach a certain **threshold**, an AP is initiated in the neuron.

3. If the threshold is reached, voltage-gated Na^+ channels remain open, causing further rapid depolarization of the membrane.
4. Na^+ entry causes the AP to reach its peak (**overshoot**), or the point at which the membrane potential is at its most positive level. At this point, voltage-gated Na^+ channels close and voltage-gated K^+ channels open to begin the return to RMP.
5. K^+ rushes out of the neuron, resulting in **repolarization**.
6. Because voltage-gated K^+ channels close more slowly than voltage-gated Na^+ channels, the membrane potential briefly becomes more negative than the RMP (ie, **hyperpolarization**).
7. As the voltage-gated K^+ channels close, RMP is restored.

In this question, voltage-gated K^+ channels were found to be *downregulated* in a patient with ALS. With fewer voltage-gated K^+ channels functioning in the patient's motor neurons, the membrane potential would return to resting levels *more slowly*, meaning **repolarization would occur more slowly**.

Therefore, it would be easier for neurons to depolarize again because the membrane potential would be *more positive* than RMP for a *longer period*. This would result in **hyperexcitability**, *not* hypoexcitability (**Choice B**).

(Choices A and C) In this patient, depolarization would be more (*not* less) likely in motor neurons, resulting in hyperexcitability, *not* hypoexcitability.

Educational objective:

During an action potential (AP), the opening and closing of voltage-gated channels determines the state of membrane polarization and propagation of the AP. If voltage-gated K^+ channels are downregulated, repolarization would occur more slowly, resulting in neuron hyperexcitability.